A CASE-STUDY BASED REPORT

ON

Box Office Revenue Prediction Using Elastic Net Regression for the Entertainment Industry

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by

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Abstract

This case study examines the prediction of movie box-office revenue using Elastic Net Regression. The project uses 1,500 movie records with budget, genre, release season, runtime, cast popularity, director experience, critic score, audience score, and marketing spend as inputs. After data cleaning, categorical encoding, and numerical scaling, the dataset is split into 1,200 training and 300 testing records. An Elastic Net model with alpha 0.1 and l1_ratio 0.5 is trained using a scikit-learn pipeline. On the test set, the model achieves an R^2 score of 0.7028, MAE of $47,143,413.08, and RMSE of $69,591,672.73. The model is saved and connected to a Flask application for interactive revenue prediction. The results demonstrate the usefulness of regularized regression for decision support, while prediction uncertainty remains important.

Introduction

Background

Movie revenue is affected by production investment, marketing, audience response, and release timing. Historical movie data can be used to estimate expected earnings using supervised regression.

Motivation for the Project

Uncertain box-office performance makes production and marketing decisions financially risky. A data-driven forecast can support early planning and resource allocation.

Overview of the Work Carried Out

The work includes dataset cleaning, exploratory analysis, encoding and scaling features, training an Elastic Net regression model, evaluating prediction errors, saving the model, and integrating it into a Flask interface.

Scope and Responsible-Use Statement

The system estimates revenue from available movie attributes; it does not guarantee commercial outcomes. Predictions should be interpreted alongside business judgment, changing market conditions, and data quality.

Problem Statement

Case-Study Scenario

A movie distributor needs an estimate of likely box-office revenue for a film based on its budget, genre, release season, runtime, popularity measures, review scores, and marketing spend.

Central Case-Study Question

How accurately can Elastic Net Regression predict movie box-office revenue from structured movie attributes?

Formal Problem Definition

Given nine predictor features X, learn a regression function f(X) that estimates continuous box-office revenue y while balancing model fit and L1/L2 regularization.

Objectives that Guide the Study

Clean the movie dataset; prepare numerical and categorical features; train Elastic Net; evaluate MAE, MSE, RMSE, and R^2; and provide a simple web-based prediction interface.

Assumptions and Limitations

The 1,500-record dataset represents the available observations. Predictions depend on the feature coverage and the train/test split. Scores do not establish causal relationships or performance on future releases.

Dataset and Exploratory Data Analysis

About the Dataset

The dataset has 1,500 movie records and 11 original columns. After removing movie_id, it contains nine input features and one target (revenue). Seven inputs are numerical: budget, runtime, cast popularity, director experience, critic score, audience score, and marketing spend. Genre and release season are categorical.

Purpose of EDA

EDA helps inspect feature distributions, detect missing data and duplicates, and understand how movie attributes relate to revenue before model fitting.

Preparation Steps Actually Performed

The movie_id field was removed and duplicate records were checked. Missing numerical entries were filled with medians and categorical entries with modes. StandardScaler

standardized numerical inputs, while OneHotEncoder encoded genre and release season with unknown-category handling.

EDA Exploration - Implementation

Python, Pandas, NumPy, Matplotlib, and Seaborn were used in the project. The provided report documents preprocessing but does not supply specific EDA plots or numerical correlation findings; therefore none are invented here.

Summary of EDA Findings

The dataset was prepared for modeling with nine predictors and revenue as the target. The supplied project materials do not report detailed distribution or correlation statistics.

Algorithm Selection and Implementation

Elastic Net combines L1 and L2 regularization and is appropriate when inputs may be correlated. The dataset was divided into 80% training (1,200 rows) and 20% testing (300 rows), with random_state=42. A scikit-learn Pipeline combined preprocessing and Elastic Net(alpha=0.1, l1_ratio=0.5). Test results: MAE = $47,143,413.08; MSE = 4,843,000,913,382,189.00; RMSE = $69,591,672.73; R^2 = 0.7028. The model was saved as movie_revenue_model.pkl and integrated with a Flask form that accepts movie information and returns predicted revenue.

Conclusion

The case study implements an end-to-end box-office revenue prediction workflow using Elastic Net Regression. The reported R^2 of 0.7028 indicates that the model explains approximately 70.28% of the observed test-set revenue variation. Prediction errors remain substantial in dollar terms, so results should be used as estimates rather than guarantees. Future work may compare additional regression algorithms, tune hyperparameters, and expand the dataset.